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Hybrid classical quantum computation for cybersecurity strategies in a layered cybersecurity model

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Abstract
Cybersecurity has become increasingly vital in our highly interconnected digital world. Many key challenges in this field can be modeled as optimization problems, involving elements such as system vulnerabilities, attack budgets, defense costs, and potential attacker gains. Recent advancements in quantum annealing have shown promise in tackling complex optimization tasks across various domains. This paper explores how quantum annealers can be leveraged to address cybersecurity problems from both offensive and defensive perspectives. For each case, a Hamiltonian is constructed in the Quadratic Unconstrained Binary Optimization (QUBO) form—suitable for input into a quantum annealer—and correctness proofs are provided. Along the way, some new QUBO construction techniques are introduced, which may be applicable to other problem areas. The theoretical findings are supported by experimental results obtained using a D-Wave simulator. The article concludes with a discussion on the practical relevance and implementation of the proposed Hamiltonians.

Keywords Cybersecurity · Quantum annealing · Cyber attack and defense

1 Introduction

Cybersecurity has become critically important in today's highly interconnected world, as it involves safeguarding sensitive information from malicious actors who seek to exploit it for harmful or damaging purposes. Many central challenges in cybersecurity can be modeled as optimization problems [1–4]. In general, such

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problems are notoriously hard to solve with both accuracy and efficiency. No universal classical method exists that can optimally solve all types of optimization problems; instead, a diverse set of specialized techniques has been developed, each effective under different conditions or for different problem instances [5]. With the emergence of quantum computing, optimization problems can now be tackled using fundamentally new approaches. Quantum annealing (QA), in particular, is designed to solve these problems by leveraging quantum tunneling—a phenomenon not available in classical computing—and by transforming an optimization problem into one of finding a minimum energy configuration. Initially introduced as a numerical algorithm [6] inspired by simulated annealing (SA) [7] and exchange Monte Carlo simulation [8], QA is especially well suited to problems involving discrete variables. A prominent example of QA's utility is in determining the ground state of spin-glass models, which map to a variety of combinatorial optimization problems, including the traveling salesman problem and satisfiability [9–11]. Today, QA is implemented on real quantum hardware known as quantum annealers, made feasible by current technology. These devices have been applied successfully in diverse domains such as portfolio management [12], protein structure prediction [13], molecular similarity analysis [14], computational biology [15], job scheduling [16], traffic optimization [17], election forecasting [18], machine learning [19–24], and autonomous vehicle routing [25]. Compared to classical simulated annealing, quantum annealing has often demonstrated superior performance [26–28], likely due to its ability to quantum tunnel through energy barriers that would trap classical algorithms.

Quantum annealing has recently gained attention as a tool for addressing cybersecurity-related challenges [29–31]. This article demonstrates how quantum annealing can be utilized to solve optimization problems arising in cybersecurity scenarios, with a particular focus on a layered security model. This model represents a security system as a rooted tree, where each edge carries a penetration cost and each node offers a prize. Cybersecurity can be analyzed from both offensive and defensive standpoints. The foundational work by Geir et al. [32] approached the problem from the attacker's perspective. In their model, an attacker begins at the root with a limited budget, traverses the tree by paying edge costs, and collects node prizes along the way, stopping once the remaining budget cannot cover further costs. The attacker's goal is to maximize the total prize collected. The authors proved that solving this problem exactly is computationally hard, but they also developed a polynomial-time approximation scheme. Expanding on this foundation, the same authors later examined the problem from a defender's viewpoint [33]. In this version, rather than being given the penetration costs and node prizes, the task is to assign them in a way that minimizes the potential reward an attacker could achieve under a fixed budget. Their findings revealed that constructing an optimal defense configuration is generally infeasible, but the study offers valuable mathematical insights into how layered defenses might be structured. More recently, Kantabutra explored the use of quantum annealing to solve specific cybersecurity optimization problems [34]. The present article builds upon and extends that work, offering a more comprehensive treatment of the approach. In this regard, the parts that appeared in [34] and the new or heavily-revised parts in this work are highlighted.

- This work and the original work share similarities in Sects. 2, 4.1.1, 4.1.3, and 5.
- The new or heavily-revised parts are Sects. 3, 4.2.1, 4.1.2, 4.2.2, 6, and 8.

2 Cybersecurity model and problems

In this section, formal definitions of the security system model and the corresponding attack and defense scenarios are presented. These definitions are based on the foundational works by Geir et al. [32, 33] and Mukdasanit and Kantabutra [35].

Definition 1 (Cybersecurity Model) A cybersecurity model M is given by a three-tuple $M = (C, P, T)$, where $T = (V, E)$ is a tree rooted at r having $n \in \mathbb{N}$ non-root vertices, C is a set of penetration costs $c_1, \ldots, c_n \in \mathbb{Z}^+$, and P is a set of prizes $p_1, \ldots, p_n \in \mathbb{Z}^+$. The attack always begins at the root r and the root always has prize 0.

Definition 2 (Security System) A security system (c, p, T) with respect to a cybersecurity model $M = (C, P, T)$ is given by two functions $c : E(T) \to C$ and $p : V(T) \setminus \{r\} \to P$. A system attack in ($c$, p, T) is given by a subtree τ of T that contains the root r of T. The cost of a system attack τ with respect to (c, p, T) is given by the cost $cst(c, \tau) = \sum_{e \in E(\tau)} c(e)$. The prize of a system attack τ with respect to (c, p, T) is given by the prize $pr(p, \tau) = \sum_{u \in V(\tau)} p(u)$. For a given budget $B \in \mathbb{Z}^+$ the maximum prize $pr^*(c, p, B)$ with respect to B is defined by $pr^*(c, p, B) = \max\{pr(p, \tau) |$ for all $\tau \subseteq T$, where $cst(c, \tau) \leq B\}$. A system attack τ whose prize is maximum with respect to a given budget B is called an optimal attack. Every system attack τ is maximal.

Generally, each system attack—referred to simply as an attack—is assumed to be maximal, reflecting the realistic goal of an attacker seeking to obtain the highest possible total prize. A few important problems in the model are formally described in the rest of this section. Problems are divided into ones which are related to the perspectives of attack and defense. In an attack on a security system an attacker may have a targeted total prize in mind and would like to know how much the budget B is necessary for such an attack to obtain the targeted total prize. More precisely, the problem definition is as follows.

Definition 3 (Minimum Budget Targeted Prize Problem)
Instance: A security system $S = (c, p, T)$ associated with M and a targeted total prize $K \in \mathbb{Z}^+$.
Instruction: Compute the minimum budget B to obtain the total prize at least K.

We now consider the inverse formulation of the Minimum Budget Targeted Prize Problem. Instead of minimizing the budget to reach a target prize, we fix a budget

and seek to maximize the total prize that can be collected. This leads to the following problem definition.

Definition 4 (Optimal Attack Optimization Problem)
Instance: A security system $S = (c, p, T)$ associated with M and a budget $B \in \mathbb{Z}^+$.
Instruction: Compute the maximum total prize $pr^*(c, p, B)$.

Next we consider some practical problems in the aspect of defense. In cybersecurity it is often useful to know the part of the system that has a particularly weak structure in a security system, making it prone to attack from malicious actors. More specifically, we are interested in the part of the security system that can be attacked by an attacker who comes with a budget B. In terms of defense, this piece of information is helpful because we can avoid keeping expensive or important items in this particular structure. The problem definition for this scenario can be defined as follows.

Definition 5 (Weak Defense Problem)
Instance: A security system $S = (c, p, T)$ associated with M and a budget $B \in \mathbb{Z}^+$.
Instruction: Compute a subsystem of S with the highest cost that can be attacked, given a budget B.

In real-world scenarios, security measures can vary in strength and may take the form of hardware, software, or organizational policies and rules. Common examples include passwords, firewalls, virtual private networks (VPNs), intrusion detection or prevention systems, antivirus programs, and role-based access control. The effectiveness of each measure can differ significantly. For instance, a short password composed solely of lowercase letters is clearly weaker than a longer one that includes uppercase letters, numbers, and special symbols. Similarly, a VPN without IP address restrictions offers less protection than one that enforces such constraints. In this article, the strongest security measure is defined as one that entirely prevents access to the part of the system it protects. Mathematically, this is modeled as removing an edge from a graph. Such a measure is represented with an infinite cost, meaning a value strictly greater than the attacker's budget. The goal is to identify a specific edge in the system where this strongest security measure should be applied in order to minimize the maximum total prize that an attacker can obtain. The defense problem is formally defined as follows.

Definition 6 (Infinity Placement Operation) Given a cybersecurity model M, let $S = (c, p, T)$ be a security system associated with M and (c', p, T) be a security system in the model after the infinity placement operation has taken place. We say that an infinity placement operation replaces an edge cost $c(e)$ with ∞ if for all edges $e' \in E(T) \setminus \{e\}$, $c'(e') = c(e')$ and $c'(e) = \infty$.

Definition 7 (Infinity Placement Optimization Problem)
Instance: A security system $S = (c, p, T)$ associated with M and a budget $B \in \mathbb{Z}^+$.

Instruction: Find an edge $e \in E(T)$ such that $pr^*(c', p, B)$ is minimized after the infinity placement operation has been performed.

The Optimal Attack Optimization Problem and the Infinity Placement Optimization Problem have been proven to be NP-hard [32] and coNP-hard [35], respectively. This suggests that it is highly unlikely for polynomial-time exact algorithms to exist for either problem unless P = NP. The most efficient known classical exact algorithms require $O(G^2 n)$ time to compute an optimal attack, and $O(G^2 n^2)$ time to identify the optimal edge for placing the strongest security measure, where $G = \sum_{p_i \in P} p_i$ represents the total prize in the system. On the approximation side, the best available classical algorithms are fully polynomial-time approximation schemes (FPTAS): a $(1-\epsilon)$-approximation with time complexity $O(\frac{1}{\epsilon^2} n^3 \log G)$ for the attack problem, and a $(1+\epsilon)$-approximation with time complexity $O(\frac{1}{\epsilon^2} n^4 \log G)$ for the edge placement problem [35]. In recent years, there has been growing interest in leveraging quantum annealing to tackle NP-hard optimization problems and others of comparable complexity [36]. At the same time, there has been considerable debate about whether quantum annealing can outperform classical algorithms in a meaningful way [36]. While most quantum computing researchers remain skeptical that quantum annealing can solve NP-complete problems in polynomial time, it is still possible that the hidden constants in its runtime could be smaller than those of classical methods. Additionally, quantum annealing benefits from quantum tunneling, which may help it find better solutions than classical algorithms [36]. Given that many problems in cybersecurity are hard to solve exactly and efficiently, it is reasonable to anticipate that quantum annealing could offer advantages for solving them.

The organization of this article is as follows. In Sect. 3, quantum annealing are introduced. Section 4 presents the main results from both the attack and defense perspectives. The first subsection is from the attack standpoint while the second subsection is from the defense standpoint. In the first subsection, given a rooted tree, a method is presented and rigorously proved for constructing a quadratic unconstrained binary optimization (QUBO) that generates all possible rooted subtrees, which form the foundation for the remaining QUBOs. Then, a scenario is considered in which an attacker targets a specific total prize from the security system and seeks to determine the corresponding required budget B. A QUBO formulation for this problem and its proof of correctness are provided. Lastly, the result related to the Optimal Attack Optimization Problem is presented through an associated and provably-correct QUBO. In the second subsection, the problem of identifying a weak subsystem within a given security system—one that could potentially be exploited by an attacker operating under a budget B—is addressed. The QUBO to solve such a problem is provided and proved correct. Finally, a hybrid algorithm is shown for solving the Infinity Placement Optimization Problem using the QUBOs described in the previous sections. The hybrid algorithm's correctness is then rigorously proved. In Sect. 5, the transformation of Higher-Order Unconstrained Binary Optimizations that are not initially in QUBO format into the QUBO format is explained, and the section concludes with tables summarizing the total number of qubits required for each QUBO. Section 6 presents experiments conducted on

a D-Wave simulator to support the theoretical findings. The article concludes with a discussion of the results and their practical implementation.

3 Quantum annealing

Interest in the idea of building a quantum computer was ignited by Feynman in a talk about using them to simulate quantum mechanical process [37]. Deutsch [38] was the first to develop a universal model of computation called the *quantum gate model* that can simulate any Turing machine computation with no more than polynomial cost overhead. *Adiabatic quantum computation* (AQC) was introduced by Farhi et al. [39] as an alternative to the gate-based model of computation. The term *quantum annealing* describes a heuristic search method to solve problems in combinatorial optimization using the adiabatic quantum computation. In AQC computation, there are neither gates nor discrete time steps. Instead, the qubit states gradually evolve according to certain forces represented by *Hamiltonians*. The AQC algorithm performs a transition from the initial Hamiltonian H_I to the final Hamiltonian H_F, and the qubit states are read at the end of the transition. Each observable n-qubit state has an associated *energy*; the *ground state* has the lowest energy over all states. According to the Adiabatic Theorem, if the qubits start in ground state and the transition is carried out slowly enough, then, with high probability, the system will finish in ground state. The goal is to design H_I such that its ground state is easy to identify, and to construct H_F so that its ground state represents the optimal solution to the given optimization problem.

AQC algorithms are designed to solve optimization problems formulated as follows: Given an *objective function* $f : D^n \to R$ defined on n binary variables $x = x_1, \ldots, x_n$ from some discrete domain D, find an assignment of values to x that minimizes $f(x)$. An AQC algorithm to solve optimization problem P with objective function $f(x)$ works on a register Q of n qubits having superposition state $|\phi_t\rangle$ at time t. The algorithm is described by a time-varying Hamiltonian $H(t)$ specified by three components:

1. An *initial Hamiltonian* H_I chosen so that the ground state of the system is easy to find.
2. A *final or problem Hamiltonian* H_F that encodes the objective function so that the ground state of Q is an eigenstate of H_F having minimum eigenvalue. In other words, a ground state s_g of Q corresponds to an optimal solution to P.
3. An *adiabatic evolution path*, a function $s(t)$ that decreases from 1 to 0 as $t : 0 \to t_f$, for some elapsed time t_f. An example is a simple linear path $s(t) = 1 - \frac{t}{t_f}$.

The Hamiltonian $H(t)$ creates a gradual transition from H_I to H_F according to the following equation.

$$H(t) = s(t)H_I + (1 - s(t))H_F \tag{1}$$

This Hamiltonian is an AQC algorithm for solving P.

The terms AQC and QA algorithms are now used interchangeably. The only main difference is that AQC is a theoretical model of quantum computation, while QA is the practical implementation of AQC principles. In this article, we refer both to the algorithm that is designed to solve a typically *NP*-hard combinatorial optimization problem. If the algorithm runs on a theoretically ideal AQC platform in an *adiabatically closed system*, which means that it experiences no interference from the environment, the algorithm is probabilistic and the Adiabatic Theorem may be used to bound the tradeoff between computation time and the probability that the system finishes in ground state. On the contrary, if the algorithm is implemented to run on a physically realized quantum platform, it is a matter of natural law that a real quantum computer must work in an *open system* that cannot perfectly be isolated from its environment. Energy from the environment creates noise that interferes with the computation and reduces the probability of finishing in ground state. Hence, the theoretical guarantees of the AQC do not necessarily apply. When no suitable performance bounds are known, we consider it to be a heuristic.

In the following sections, how the quantum annealing can be used to solve some optimization problems in a cybersecurity model is discussed. The focus will be simply on deriving a problem Hamiltonian H_F that encodes the objective function in the form of quadratic unconstrained binary optimization (QUBO) so that the ground state of Q is an eigenstate of H_F having a minimum eigenvalue. We note here that this QUBO form is more convenient to work with, partly because heuristic algorithms are typically implemented to return binary-valued solutions and that this QUBO form can be translated back to the equivalent Ising formula readily. Figure 1 illustrates general steps in converting a problem into a QUBO that is ready for a quantum annealer.

4 Contributions

In this section, all of our contributions are discussed. We show how to construct QUBOs for several interesting problems in the context of cybersecurity. All these constructions do not necessarily conform to a QUBO format at first and later a standard way to modify each of them into an equivalently correct QUBO format will be discussed. Because our cybersecurity model is based on graphs, there are a few existing graph-based QUBO constructions that are similar in ideas to the ones in this section [36, 40–42]. However, unlike QUBOs in this section that are proved rigorously correct, all of them are experimental.

4.1 Attack problems

4.1.1 QUBO for rooted subtrees

Because an attack in a security system in the model is defined as a rooted subtree, a QUBO that generates legitimate rooted subtrees is needed. Given a rooted tree T, our goal is to generate all possible rooted subtrees of T. Let q_u and q_v be two binary variables representing an edge $\{u, v\} \in E(T)$ where u is the vertex closer to the root. Here, we assume that the tree T is rooted, and each vertex is assigned

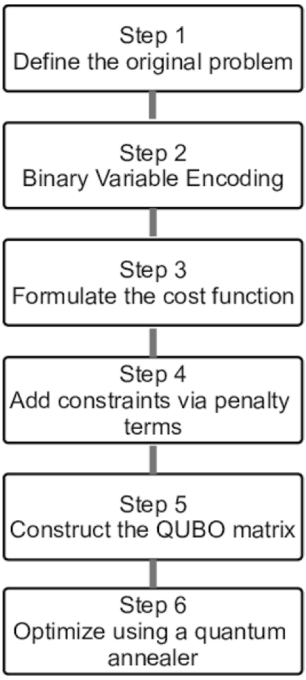


Fig. 1 Steps in QUBO transformation

a level (tier) based on its distance from the root. For an edge $\{u, v\} \in E(T)$, the vertex with the smaller tier number is considered to be closer to the root. Both q_u and q_v are each set the values according to Table 1 so that vertices u and v are part of a rooted subtree of T. Hence, we would like to have all rooted subtrees of 1's. More precisely, T' is a rooted subtree of T if and only if for each path $u_0u_1 \dots u_l$ in T and $u_0 = r$ there exists $0 \le k \le l$ such that $u_0, u_1, \dots, u_k \in V(T')$ and $u_0u_1 \dots u_k$ forms a path in T' and (possibly empty list) $u_{k+1}, u_{k+2}, \dots, u_l \notin V(T')$. A rooted subtree T' of 1's is where $q_{u_0} = q_{u_1} = \dots = q_{u_k} = 1$ and $q_{u_{k+1}} = q_{u_{k+2}} = \dots = q_{u_l} = 0$ for each such a path. From this point on, we call a rooted subtree T' of 1's simply a rooted subtree T' when there is no confusion.

Consider an edge $\{u, v\} \in E(T)$ and the behavior of the binary variables in Table 1. Using the most general form of a two binary variables QUBO $H_{uv} = aq_u + bq_v + cq_uq_v$ and solving this equation, we have $H_{uv} = q_v - q_uq_v$ for representing the behavior of the edge $\{u, v\}$. To create a QUBO for each edge in the given tree, we sum H_{uv} over all edges in $E(T)$.

Table 1 Two binary variables

q_u	q_v	H_{uv}
0	0	0
0	1	1
1	0	0
1	1	0

Lemma 1 *Let* $H_A = \sum_{\{u,v\}\in E} \mathrm{H}_{uv} = \sum_{\{u,v\}\in E} (q_v - q_u q_v)$. *The QUBO* H_A *is* 0 *if and only if all edges* $\{u, v\} \in E(T)$ *are either* $q_u = 0$ *and* $q_v = 0$, $q_u = 1$ *and* $q_v = 0$, *or* $q_u = 1$ *and* $q_v = 1$.

Proof (→) Suppose $H_A = \sum_{\{uv\}\in E} \mathrm{H}_{uv} = 0$, implying that, for each edge $\{u, v\} \in E$, $q_v - q_u q_v = 0$. According to Table 1, for each edge $\{u, v\} \in E(T)$ only the cases when $q_u = 0$ and $q_v = 0$, $q_u = 1$ and $q_v = 0$, or $q_u = 1$ and $q_v = 1$ produce 0. So, the first direction holds.

(←) Suppose all edges $\{u, v\} \in E(T)$ are either $q_u = 0$ and $q_v = 0$, $q_u = 1$ and $q_v = 0$, or $q_u = 1$ and $q_v = 1$. Substitute values of q_u and q_v in each case in $q_v - q_u q_v$. Each of these cases produces 0 and therefore $H_A = 0$. Both directions hold. Therefore, the lemma is correct. □

At this point all edges $\{u, v\}$ with $q_u = 0$ and $q_v = 1$ are filtered out. We now need to connect the remaining edges to form rooted subtrees of 1's. We achieve this by adding $H_B = 1 - q_0$, where q_0 is the binary variable corresponding to the root vertex r of T. Note that $H_B = 0$ if and only if $q_0 = 1$.

Lemma 2 *Let* $H_A = \sum_{\{uv\}\in E} \mathrm{H}_{uv} = \sum_{\{uv\}\in E} (q_v - q_u q_v)$ *and* $H_B = 1 - q_0$ *and let* $P = (q_0, q_1, \ldots, q_d, q_{d+1}, \ldots, q_k, q_{k+1})$ *be a simple path from the root* q_0 *to a leaf* q_{k+1} *such that the edges are* $\{q_i, q_{i+1}\}$ *where* $i = 0, 1, 2, 3, \ldots, k$, *where* h *is the height of the tree* T *and* $1 \le k \le h$ *and* $0 \le d \le h$. *The QUBO* $H_A + H_B = 0$ *if and only if, for all* $d \ge 0$, *there exists a bit sequence* $q_0 = q_1 = \ldots = q_d = 1$ *and* $q_{d+1} = q_{d+2} = \ldots = q_{k+1} = 0$ *for each such path* P.

Proof (→) Suppose $H_A + H_B = 0$. We prove by induction on d. Consider the base case $d = 0$. Because $H_B = 0$, the root binary variable q_0 is set to 1. Because $H_A = 0$, q_1 can be 0 and q_2 to q_{k+1} can also be 0 according to Lemma 1. Hence, there exists a bit sequence $q_0 = 1$ and $q_1 = q_2 = \ldots = q_{k+1} = 0$ for each such path P and the base case holds.

Suppose for $d \ge 0$, there exists a bit sequence $q_0 = q_1 = \ldots = q_d = 1$ and $q_{d+1} = q_{d+2} = \ldots = q_{k+1} = 0$ for such a path P. We show that this claim also holds for $d + 1$. That is, there exists a bit sequence $q_0 = q_1 = \ldots = q_d = q_{d+1} = 1$ and $q_{d+2} = q_{d+3} = \ldots = q_{k+1} = 0$ for such a path P. By the supposition, there exists a bit sequence $q_0 = q_1 = \ldots = q_d = 1$ and $q_{d+1} = q_{d+2} = \ldots = q_{k+1} = 0$ for such a path P. Reset q_{d+1} in this path to 1. By Lemma 1, this single change from $q_d = 1$ and $q_{d+1} = 0$ to $q_d = 1$ and $q_{d+1} = 1$ still maintains $H_A + H_B = 0$. Hence, the claim holds for $d + 1$.

(←) Suppose for all $d \ge 0$, there exists a bit sequence $q_0 = q_1 = \ldots = q_d = 1$ and $q_{d+1} = q_{d+2} = \ldots = q_{k+1} = 0$ for each such path P. In this particular, path P there are only three possible cases of bit assignments for an edge $\{q_u, q_v\}$. Consider the first case where $q_u = 1$ and $q_v = 1$. By Lemma 1, this case produces 0. The cases where $q_u = 1$ and $q_v = 0$ and $q_u = 0$ and $q_v = 0$ also produce 0 by Lemma 1. Therefore, $H_A + H_B = 0$.

Therefore, the lemma holds. □

We say that a configuration $\{q_v\}$ satisfies $H_A + H_B = 0$ if and only if restricting T to edges $\{\{u, v\} \in E(T) | q_u = q_v = 1\}$ produces a rooted subtree T' of T. In other words, we are interested in a subtree T' of T whose q_v's are all set to 1.

4.1.2 QUBO for an attacking budget

In this section, we construct the QUBO to solve the Minimum Budget Targeted Prize Problem according to Definition 3. Let $AP = \sum_{v \in V(T)} p(v)$ and $H_F = (1 - \sum_{i=0}^{AP-K} r_i)^2 + (K + \sum_{i=0}^{AP-K} ir_i - \sum_{v \in V(T)} p_v q_v)^2$, where r_i is a binary variable, and $H_G = \sum_{\{u,v\} \in E(T)} c_{uv} q_u q_v$. Let $AC = \sum_{\{u,v\} \in E(T)} c_{uv}$ and $H_C = (1 - \sum_{j=1}^{AC} z_j)^2 + (\sum_{j=1}^{AC} jz_j - \sum_{\{u,v\} \in E(T)} c_{uv} q_u q_v)^2$ be the QUBO representing the total cost. Finally, let H_{MB} be the following QUBO.

$$
\begin{aligned}
H_{MB} &= H_A + H_B + H_C + H_F + H_G \\
&= \sum_{\{u,v\} \in E} (q_v - q_u q_v) + 1 - q_0 \\
&\quad + \left(1 - \sum_{j=1}^{AC} z_j\right)^2 + \left(\sum_{j=1}^{AC} jz_j - \sum_{\{u,v\} \in E(T)} c_{uv} q_u q_v\right)^2 \\
&\quad + \left(1 - \sum_{i=0}^{AP-K} r_i\right)^2 + \left(K + \sum_{i=0}^{AP-K} ir_i - \sum_{v \in V(T)} p_v q_v\right)^2 \\
&\quad + \sum_{\{u,v\} \in E(T)} c_{uv} q_u q_v
\end{aligned}
$$

Theorem 1 *Let $H_{MB} = L(H_A + H_B + H_C + H_F) + H_G$, where L is a constant such that $L > c_{\max}|E(T)|$, where $c_{\max}$ is the maximum value in C. Given a targeted total prize K, H_{MB} yields the minimum budget $B = j$ to obtain the total prize at least K at the end of the anneal, assuming the ideal annealing process.*

Proof $H_A + H_B + H_C = 0$ if and only if at the lowest energy state the annealing process produces a rooted subtree T' of T with a corresponding total cost jz_j by Lemma 4. Additionally, $H_F = (1 - \sum_{i=0}^{AP-K} r_i)^2 + (K + \sum_{i=0}^{AP-K} ir_i - \sum_{v \in V(T)} p_v q_v)^2 = 0$ if and only if there exists a single r_i such that $K + ir_i = \sum_{v \in V(T)} p_v q_v$ at the lowest energy state, which implies $K \leq \sum_{v \in V(T')} p_v q_v$. Hence, $H_A + H_B + H_C + H_F = 0$ if and only if at the lowest energy state the annealing process produces a rooted subtree T' of T with a corresponding total cost jz_j and there exists a rooted subtree T' of a total prize at least K. Because the annealing process terminates at the lowest energy state in the ideal environment, $H_{MB} = L(H_A + H_B + H_C + H_F) + H_G$ terminates when H_{MB} has the lowest value, which implies H_G has the lowest value since $H_A + H_B + H_C + H_F = 0$. Therefore, in the end of the anneal, H_{MB} yields the minimum budget $B = \sum_{\{u,v\} \in E(T)} c_{uv} q_u q_v = j$ to obtain the total prize at least K. □

Observe again that L is a penalty that occurs when $H_A + H_B + H_C + H_F$ does not equal 0, implying that we solve another different problem. We would like to avoid this situation.

4.1.3 QUBO for the optimal attack

In a security system $S = (c, p, T)$ associated with M, an attacker can come with a certain budget B. From the attacker's perspective, we would like to find a way to attack the security system and gain the maximum total prize within the given budget B. This problem was formally described in Definition 4. We begin designing our QUBO for this problem by extending $H_A + H_B$ to cover both costs and prizes of each rooted subtree. By Definition 2, each vertex $v \in V(T)$ is associated with prize $p_v = p(v)$ and each edge $\{u, v\} \in E(T)$ is associated with cost $c_{uv} = c(\{u, v\})$. Furthermore, let y_i and z_j be binary variables for picking the values i, j for a total prize and a total cost respectively and $q_v = 1$ if and only if $v \in V(T)$ is chosen to be in a rooted subtree. Lastly, let $AP = \sum_{v \in V(T)} p(v)$ and $AC = \sum_{\{u,v\} \in E(T)} c_{uv}$ and let H_P and H_C be the QUBOs representing the total prize and the total cost, respectively.

$$H_P = \left(1 - \sum_{i=1}^{AP} y_i\right)^2 + \left(\sum_{i=1}^{AP} iy_i - \sum_{v \in V(T)} p_v q_v\right)^2$$

$$H_C = \left(1 - \sum_{j=1}^{AC} z_j\right)^2 + \left(\sum_{j=1}^{AC} jz_j - \sum_{\{u,v\} \in E(T)} c_{uv} q_u q_v\right)^2$$

Observe that $H_P = 0$ is forced to have only a single total prize and similarly $H_C = 0$ is forced to have only a single total cost. We combine both together by setting $H_{PC} = H_P + H_C$.

Lemma 3 *$H_A + H_B + H_{\mathrm{PC}} = 0$ if and only if the annealing process produces a rooted subtree T' of T with a corresponding total prize iy_i and a corresponding total cost jz_j.*

Proof ($\rightarrow$) Suppose $H_A + H_B + H_{\mathrm{PC}} = 0$. By Lemma 2, a configuration $\{q_v\}$ satisfies $H_A + H_B = 0$ if and only if restricting T to edges $\{\{u, v\} \in E(T) | q_u = q_v = 1\}$ produces a rooted subtree T' of T. For all q_v's in the configuration $\{q_v\}$ of T', $q_u = 1$ and $q_v = 1$ contributes p_v to the total prize and c_{uv} to the total cost in H_P and H_C, respectively. $H_P = 0$ implies that there is only one $y_i = 1$ in the first term and this y_i sets $iy_i = \sum_{v \in V(T)} p_v q_v$ in the second term. Similarly, $H_C = 0$ implies that there is only one $z_j = 1$ in the first term and this z_j sets $jz_j = \sum_{\{u,v\} \in E(T)} c_{uv} q_u q_v$ in the second term. Hence, the first direction holds.

($\leftarrow$) Suppose the annealing process produces a rooted subtree T' of T with a corresponding total prize iy_i and a corresponding total cost jz_j. By Lemma 2, a configuration $\{q_v\}$ satisfies $H_A + H_B = 0$ if and only if restricting T to edges $\{\{u, v\} \in E(T) | q_u = q_v = 1\}$ produces a rooted subtree T' of T. Hence, $H_A + H_B = 0$.

Because the annealing process always terminates at the lowest energy state, in the first term of H_P, there is only one $y_i = 1$ and in the second term $iy_i = \sum_{v \in V(T)} p_v q_v$. Hence, $H_P = 0$. Similarly, because the annealing process always terminates at the lowest energy state, in the first term of H_C, there is only one $z_j = 1$ and in the second term $jz_j = \sum_{\{u,v\} \in E(T)} c_{uv} q_u q_v$. Hence, $H_C = 0$ and $H_{PC} = 0$. Then $H_A + H_B + H_{PC} = 0$. The lemma is true. □

We say that a configuration $\{q_v\}, \{y_i\}, \{z_j\}$ satisfies $H_A + H_B + H_{PC} = 0$ if and only if the subtree T' determined by $\{q_v\}$ has a total prize i and a total cost j and $y_i = z_j = 1$.

We next want to maximize the total prize and make sure that the total cost does not exceed a given budget B. To maximize the total prize, we create the following QUBO H_D.

$$H_D = -\sum_{v \in V(T)} p_v q_v$$

Observe that by making it negative, we are turning a maximization problem into a minimization problem. To make sure that the total cost is no greater than a given budget B, H_E is reused.

$$H_E = \left(1 - \sum_{i=0}^{AC} b_i\right)^2 + \left(\sum_{\{u,v\} \in E(T)} c_{uv} q_u q_v + \sum_{i=0}^{AC} i b_i - B\right)^2$$

Again observe that $H_E = 0$ if and only if exactly one b_i in the first term is set to be 1 and $\sum_{\{u,v\} \in E(T)} c_{uv} q_u q_v + i b_i = B$ and that $H_E \geq 0$. Let H_{attack} be as follows.

$$\begin{aligned}
H_{attack} &= H_A + H_B + H_{PC} + H_D + H_E \\
&= \sum_{\{u,v\} \in E} (q_v - q_u q_v) + 1 - q_0 + \left(1 - \sum_{i=1}^{AP} y_i\right)^2 \\
&\quad + \left(\sum_{i=1}^{AP} i y_i - \sum_{v \in V(T)} p_v q_v\right)^2 + \left(1 - \sum_{j=1}^{AC} z_j\right)^2 \\
&\quad + \left(\sum_{j=1}^{AC} j z_j - \sum_{\{u,v\} \in E(T)} c_{uv} q_u q_v\right)^2 \\
&\quad - \sum_{v \in V(T)} p_v q_v + \left(1 - \sum_{i=0}^{AC} b_i\right)^2 \\
&\quad + \left(\sum_{\{u,v\} \in E(T)} c_{uv} q_u q_v + \sum_{i=0}^{AC} i b_i - B\right)^2
\end{aligned}$$

We remark here that $H_A + H_B + H_{PC} + H_E$ is at least 0 for all solutions.

Theorem 2 *Let* $H_{\text{attack}} = L(H_A + H_B + H_{PC} + H_E) + H_D$, *where* $L = p_{\max}|V|$ *is a constant.* H_{attack} *yields* $-1\times$ *an optimal solution to the* Optimal Attack Optimization Problem *at the end of the anneal, assuming the ideal annealing process.*

Proof Because $H_A + H_B + H_{\text{PC}} + H_E$ is at least 0 for all solutions, $L(H_A + H_B + H_{\text{PC}} + H_E)$ produces a solution equal to 0 at the end of the anneal. By adding H_D to maximize the total prize, $L(H_A + H_B + H_{\text{PC}} + H_E) + H_D$ gives a minimum energy solution that is $-1\times$ the maximum total prize with the total cost at most B at the end of the anneal. □

Observe that if a constraint in $H_A + H_B + H_{\text{PC}} + H_E$ is violated, a solution obtained is no longer a solution to the Optimal Attack Optimization Problem. However, if a constraint in H_D is violated, i.e., some p_v should have been included but is not, and no constraint in $H_A + H_B + H_{\text{PC}} + H_E$ is violated, a solution obtained is a feasible solution to the Optimal Attack Optimization Problem. Therefore, for each violation of a constraint in $L(H_A + H_B + H_{\text{PC}} + H_E)$, a penalty of at least $L = p_{\max}|V|$ occurs. Because $L \geq \sum_{v \in V(T)} p_v q_v$, each constraint in $L(H_A + H_B + H_{\text{PC}} + H_E)$ is less preferable to be violated because the energy penalty is higher.

4.2 Defense problems

4.2.1 QUBO for weak defense

In this section, we describe the QUBO that discovers a subsystem in the Weak Defense Problem according to Definition 5 as follows. By Definition 2, each edge $\{u, v\} \in E(T)$ is associated with cost $c_{uv} = c(\{u, v\})$. Given a budget B, we want to find a subsystem that has a sum of edge costs at most B. Furthermore, let z_j be a binary variable for picking the values j for a total cost and $q_v = 1$ if and only if $v \in V(T)$ is chosen to be in a rooted subtree. Let $AC = \sum_{\{u,v\} \in E(T)} c_{uv}$ and let H_C be the QUBO representing the total cost of a rooted subtree and is defined as follows.

$$H_C = \left(1 - \sum_{j=1}^{AC} z_j\right)^2 + \left(\sum_{j=1}^{AC} j z_j - \sum_{\{u,v\} \in E(T)} c_{uv} q_u q_v\right)^2$$

Observe that $H_C = 0$ is forced to have only a single total cost. Furthermore, let $H_S = -\sum_{\{u,v\} \in E(T)} c_{uv} q_u q_v$ be a QUBO to maximize the total cost and $H_E = (1 - \sum_{i=0}^{AC} b_i)^2 + (\sum_{\{u,v\} \in E(T)} c_{uv} q_u q_v + \sum_{i=0}^{AC} i b_i - B)^2$ be a QUBO to select the appropriate cost no greater than B. We incorporate all these QUBOs into QUBO H_W as follows.

$$
\begin{aligned}
H_W &= H_A + H_B + H_C + H_S + H_E \\
&= \sum_{\{u,v\}\in E} (q_v - q_u q_v) + 1 - q_0 \\
&\quad + \left(1 - \sum_{j=1}^{AC} z_j\right)^2 + \left(\sum_{j=1}^{AC} j z_j - \sum_{\{u,v\}\in E(T)} c_{uv} q_u q_v\right)^2 \\
&\quad - \sum_{\{u,v\}\in E(T)} c_{uv} q_u q_v + \left(1 - \sum_{i=0}^{AC} b_i\right)^2 \\
&\quad + \left(\sum_{\{u,v\}\in E(T)} c_{uv} q_u q_v + \sum_{i=0}^{AC} i b_i - B\right)^2
\end{aligned}
$$

Lemma 4 *$H_A + H_B + H_C = 0$ if and only if at the lowest energy state the annealing process produces a rooted subtree T' of T with a configuration $\{q_v\}$ that satisfies $H_A + H_B = 0$ and a corresponding total cost jz_j for some j.*

Proof ($\rightarrow$) Suppose $H_A + H_B + H_C = 0$. By Lemma 2, $H_A + H_B = 0$ implies that there exists a bit sequence $q_0 = q_1 = \ldots = q_d = 1$ and $q_{d+1} = q_{d+2} = \ldots = q_{k+1} = 0$ for each such path from the root binary variable q_0. For each path $q_0 = q_1 = \ldots = q_d = 1$, each pair of $q_u = 1$ and $q_v = 1$ corresponding to an edge $\{u, v\}$ in the subtree T' contributes the cost c_{uv} to $\sum_{\{u,v\}\in E(T)} c_{uv} q_u q_v$. $H_C = 0$ implies that there is only one $z_j = 1$ in the first term and this z_j sets $jz_j = \sum_{\{u,v\}\in E(T)} c_{uv} q_u q_v$ in the second term. Hence, the first direction holds.

($\leftarrow$) Suppose at the lowest energy state the annealing process produces a rooted subtree T' of T with a configuration $\{q_v\}$ that satisfies $H_A + H_B = 0$ and a corresponding total cost jz_j for some j. We only need to show that $H_C = 0$. By our supposition, we only have the one cost jz_j and therefore, in the first term of H_C, z_j must be equal to 1 to obtain the lowest energy state 0. Furthermore, this jz_j must be equal to the cost of the subtree $T' = \sum_{\{u,v\}\in E(T)} c_{uv} q_u q_v$ to obtain the lowest energy state 0 in the second term of H_C. Hence, $H_C = 0$. Then $H_A + H_B + H_C = 0$. The lemma is true. □

Theorem 3 *Let $H_W = L(H_A + H_B + H_C + H_E) + H_S$, where L is a constant such that $L > c_{max}|E(T)|$, where c_{max} is the maximum value in C. Given a budget B, H_W yields an optimal solution with the highest total cost at the end of the anneal, assuming the ideal annealing process.*

Proof Observe that $H_E = (1 - \sum_{i=0}^{AC} b_i)^2 + (\sum_{\{u,v\}\in E(T)} c_{uv} q_u q_v + \sum_{i=0}^{AC} i b_i - B)^2 = 0$ if and only if the annealing process is at the lowest energy state and the total cost does not exceed the budget B. By Lemma 4, $H_A + H_B + H_C + H_E = 0$ if and only if the annealing process is at the lowest energy state and produces a rooted subtree T' of T with a corresponding total cost jz_j and the total cost does not exceed the budget B. When the annealing process terminates at the lowest energy state in the ideal

condition, $L(H_A + H_B + H_C + H_E) + H_S$ equals the lowest value, implying H_S equals the highest total cost at the end of the anneal. □

Observe that L is a penalty that occurs when $H_A + H_B + H_C + H_E$ does not equal 0, implying that we solve another different problem. We would like to avoid this situation.

4.2.2 QUBO for the optimal defense

In this section, a solution to the Infinity Placement Optimization Problem according to Definition 7 is discussed. Recall that we are given a security system $S = (c, p, T)$ associated to a cybersecurity model M and we would like to find an edge $e \in E(T)$ that minimizes the maximum total prize from an attack after the infinity cost has been placed on the edge e. The idea behind our QUBO design is to reuse H_{attack} in Sect. 4.1.3 d times with d different placements of infinity, where d is the degree of the root vertex of T. We would then like to identify the attack with the minimized maximum total prize within a given budget so that we know where to place the corresponding infinity cost. First, we need to have a QUBO for *exactly one* placement of infinity.

Lemma 5 *Let s_i be a binary variable. For any integer $d \geq 1$, $H_d^\infty = -\sum_{i=1}^{d} s_i + 2\sum_{i=1}^{d-1}\sum_{j>i}^{d} s_i s_j + 1 = 0$ if and only if exactly one s_i is set to 1.*

Proof It is clear that if exactly one s_i is set to 1 then $H_d^\infty = 0$. We prove the other direction by mathematical induction on d. Consider the case $d = 1$. $H_d^\infty = -\sum_{i=1}^{1} s_i + 1 = 0$ clearly implies that $s_1 = 1$. The base case holds. Suppose if $H_d^\infty = -\sum_{i=1}^{d} s_i + 2\sum_{i=1}^{d-1}\sum_{j>i}^{d} s_i s_j + 1 = 0$, then exactly one s_i is set to 1. We will show that if $H_{d+1}^\infty = -\sum_{i=1}^{d+1} s_i + 2\sum_{i=1}^{d}\sum_{j>i}^{d+1} s_i s_j + 1 = 0$, then exactly one s_i is set to 1. Consider $H_{d+1}^\infty = 0$. We expand it to $H_{d+1}^\infty = -\sum_{i=1}^{d} s_i - s_{d+1} + 2\sum_{i=1}^{d-1}\sum_{j>i}^{d} s_i s_j + 1 + 2\sum_{i=1}^{d} s_i s_{d+1} = 0$. There are two cases to consider. If $s_{d+1} = 0$, all new terms with $d + 1$ are summed to 0. The inductive hypothesis applies, $H_{d+1}^\infty = H_d^\infty + 0 = 0$ and exactly one $s_i = 1$. Consider when $s_{d+1} = 1$ and $H_{d+1}^\infty = 0$. We will show by contradiction that there is exactly s_{d+1} that is set to 1. Suppose there are two or more s_i's, $1 \leq i \leq d + 1$, that are set to 1. The term $-s_{d+1} = -1$ and cancels the term $+1$ in H_{d+1}^∞. Now consider the first term $-\sum_{i=1}^{d} s_i$ and the last term $2\sum_{i=1}^{d} s_i s_{d+1}$ in H_{d+1}^∞. For each $s_i = 1$ in the first term, we have a contribution of -1 from the first term and 2 from the last term. Hence, $H_{d+1}^\infty \geq 1$ because the term $2\sum_{i=1}^{d-1}\sum_{j>i}^{d} s_i s_j$ is always positive. $H_{d+1}^\infty \neq 0$ is a contradiction. Both cases hold. The lemma is correct. □

At this point, we have a mechanism that sets exactly one s_i to 1 if and only if $H_d^\infty = 0$. We can incorporate H_d^∞ into H_{attack} to produce a QUBO for the optimal defense. The hybrid algorithm is now to eliminate, one by one, the placement of the infinity cost that we know *for certain* that it is not an optimal solution. Eventually,

the last remaining placement of the infinity cost becomes an optimal solution that minimizes the maximum total prize from an attack with a budget B.

In terms of implementation, let the infinity cost be any value *strictly* greater than a given budget B. We have a few observations. First, the value for the infinity cost can be defined because the budget B is given as part of the input instance. Second, the optimal edge for the infinity placement is always the one among all d edges adjacent to the root vertex r for otherwise we can always move infinity upward to reduce a total prize from an attack. Third, this problem is to determine a *minimum* value among all d *maximum* total prizes. It is a combination of both minimization and maximization in one single problem. It is not obvious how one can directly formulate a QUBO for it. Our approach depends on a hybrid algorithm, which will be described shortly.

We will create a QUBO called H_{defense}. Among d placements of the infinity cost, one of such placements which gives the minimum of all d total prizes from any attack will be the *best defense*.

Suppose that the d edges are numbered increasingly from left to right. In the preprocessing phase of the hybrid algorithm, we pick an integer $1 \le l \le d$ uniformly at random. Let H^l_{attack} be the QUBO with the initial tree T such that the infinity cost is placed on the edge l^{th} adjacent to the root vertex r in the tree. We use this H^l_{attack} to produce an optimal solution to the Optimal Attack Optimization Problem according to the Theorem 2. We will use this optimal solution as a threshold U in the following construction for the QUBO H_{defense}. Observe that the best defense gives the solution that is at most U.

Let $T_1 < T_2$ be constants and $H_{\text{defense}} = \sum_{i=1}^{d-1} T_1(s_i H^i_{\text{attack}}) + T_2(H^\infty_d)$ such that H^i_{attack} is the modified QUBO for the optimal attack on the input instance in which the infinity cost is placed on the edge i adjacent to the root vertex r and each H^i_{attack}, $1 \le i \le d$, $i \ne l$, can be defined as follows. Let $H_D^{\infty,i}$ be $U - \sum_{v \in V(T)} p_v q_v$ where the infinity cost is placed on the corresponding edge i.

$$
\begin{aligned}
H^i_{attack} &= H_A + H_B + H_{PC} + H_D^{\infty,i} + H_E \\
&= \sum_{\{u,v\}\in E} (q_v - q_u q_v) + 1 - q_0 + (1 - \sum_{i=1}^{AP} y_i)^2 \\
&\quad + \left(\sum_{i=1}^{AP} i y_i - \sum_{v\in V(T)} p_v q_v\right)^2 + \left(1 - \sum_{j=1}^{AC} z_j\right)^2 \\
&\quad + \left(\sum_{j=1}^{AC} j z_j - \sum_{\{u,v\}\in E(T)} c_{uv} q_u q_v\right)^2 \\
&\quad + \left(U - \sum_{v\in V(T)} p_v q_v\right) + \left(1 - \sum_{i=0}^{AC} b_i\right)^2 \\
&\quad + \left(\sum_{\{u,v\}\in E(T)} c_{uv} q_u q_v + \sum_{i=0}^{AC} i b_i - B\right)^2
\end{aligned}
$$

A few remarks are in order. First, $H_A + H_B + H_{\text{PC}} + H_E$ is at least 0 for all solutions and exactly 0 for legitimate solutions. Second, $T_1 < T_2$ because if H_d^{∞} is not 0, the resulting solution will not be a feasible solution for the Infinity Placement Optimization Problem. Third, H^i_{attack} is essentially the same as H_{attack} in Theorem 2 except the part of $U - \sum_{v \in V(T)} p_v q_v$.

Algorithm 1 is a classical-quantum algorithm used to find the edge for the infinity cost placement that yields the minimum total prize from an attack with a budget B. In this algorithm, let i^- be the edge i where the infinity cost has been placed and this placement causes H_{defense} to become negative. Likewise, let i^+ be the edge i where the infinity cost has been placed and this placement causes H_{defense} to become positive. Using these notations, the hybrid algorithm can be described as follows.

Algorithm 1 Hybrid Algorithm(H_{defense})

```
1:  answer = l;
2:  repeat
3:      execute the quantum annealing process with H_defense;
4:      if the resulting energy solution from H_defense is negative then
5:          remove the corresponding H^{i-}_attack from H_defense;
6:      end if
7:      if the resulting energy solution from H_defense is positive then
8:          answer = i^+;
9:          U = U - the energy solution from H_defense;
10:         remove the H^{i+}_attack from H_defense;
11:         replace old U with new U in each of the remaining H^i_attack in H_defense;
12:     end if
13: until no more H^i_attack is left;
14: return answer;
```

Theorem 4 (Algorithm's Correctness) *Assume the ideal annealing condition. Given H_{defense}, the hybrid algorithm gives the placement of the infinity cost on the edge answer that yields the least total prize from an attack at the end of the anneal.*

Proof Let $T_1 < T_2$ be constants and $H_{\text{defense}} = \sum_{i=1}^{d-1} T_1(s_i H^i_{\text{attack}}) + T_2(H_d^{\infty})$ such that H^i_{attack}, $1 \le i \le d$, $i \ne l$. We will prove this theorem by induction on the following loop invariant $P(k)$, $1 \le k \le d-1$. Let $P(k)$ be the statement that at iteration k of the Hybrid Algorithm($H_{defense}$) the variable *answer* is the edge with the infinity placement that yields the least total prize from an attack among the $k+1$ edges.

Before the loop is entered, the loop invariant holds because at iteration 0 of the Hybrid Algorithm(H_{defense}) the variable *answer* $= l$ (line 1) is the edge with the infinity placement that yields the least total prize from an attack among the 1 edge.

Suppose the loop invariant holds after k iterations. There are two cases to consider in iteration $k+1$. If the resulting energy solution from H_{defense} in line 4 is lowest and negative, exactly one s_{i^-} in H_{defense} is set to 1 by Lemma 5. By Lemma 5, there is only one $H^{i^-}_{\text{attack}}$ with the minimum energy from the term H_D^{∞,i^-} at the end of

the anneal. Because $H_D^{\infty,i^-} = U - \sum_{v \in V(T)} p_v q_v$ where the infinity cost is placed on the corresponding edge i^-, the total prize from this particular placement of the infinity cost is higher than U, which is the total prize from placing the infinity cost on the edge *answer*. This edge i^- is not the optimal placement of the infinity cost. The edge *answer* remains the optimal placement among the $k+1$ placements and in line 5 the algorithm accordingly removes this $H_{\text{attack}}^{i^-}$ from H_{defense}. If the resulting energy solution from H_{defense} in line 7 is lowest and positive, exactly one s_{i^+} in H_{defense} is set to 1 by Lemma 5. By Lemma 5, there is only one $H_{\text{attack}}^{i^+}$ with the minimum energy from the term H_D^{∞,i^+} at the end of the anneal. Because $H_D^{\infty,i^+} = U - \sum_{v \in V(T)} p_v q_v$, where the infinity cost is placed on the corresponding edge i^+, is positive, the total prize from this particular placement of the infinity cost on edge i^+ is at most U, which is the total prize from placing the infinity cost on the edge *answer*. Hence, the edge *answer* that gives U is not the optimal placement of the infinity cost in iteration $k+1$ but the edge i^+ is. The variable *answer* is set to i^+ accordingly on line 8. Lines 9–11 reset the constant U to the total energy of $\sum_{v \in V(T)} p_v q_v$ in which the infinity cost is placed on the edge i^+. Hence, in both cases, at iteration $k+1$ of the Hybrid Algorithm($H_{defense}$) the variable *answer* is the edge with the infinity placement that yields the least total prize from an attack among the $k+1$ edges.

Therefore, at iteration $k = d-1$ of the Hybrid Algorithm($H_{defense}$), the variable *answer* is the edge with the infinity placement that yields the least total prize from an attack among the d edges. Hence, $P(K)$ is true and the theorem holds. □

5 QUBO transformations

We are now ready to discuss a way to turn H_{defense} into a corresponding QUBO form. A standard and widely used technique called *polynomial reduction by minimum selection* will be applied. The proof and details of this technique appears in [43]. In addition, the dimod package in the D-Wave Ocean software that uses this technique provides classes and functions for constructing and manipulating BQMs, which are essential for representing problems in QUBO form [44]. We borrow this technique and discuss it here.

To reformulate a non-quadratic higher-degree polynomial to QUBO, substitute a term in the form of $as_iq_uq_v$, where a is a real number, with one of the following quadratic terms, depending on whether a is negative or positive. Let w be an ancillary binary variable. If $a < 0$, let $as_iq_uq_v = aw(s_i + q_u + q_v - 2)$. If $a > 0$, let $as_iq_uq_v = a\{w(s_i + q_u + q_v - 1) + (s_iq_u + q_uq_v + q_vs_i) - (s_i + q_u + q_v) + 1\}$. This technique exploits the identities $s_iq_uq_v = \max_w\{w(s_i + q_u + q_v - 2)\}$ for the case $a < 0$ and $s_iq_uq_v = \min_w\{w(s_i + q_u + q_v - 1) + (s_iq_u + q_uq_v + q_vs_i) - (s_i + q_u + q_v) + 1\}$ for the case $a > 0$. Tables 2 and 3 illustrate the transformation from a triplet of variables to an equivalent QUBO form when $a < 0$ and $a > 0$, respectively. We repeatedly substitute such terms until no such term is left.

Table 2 Triplet transformation into QUBO when $a < 0$

s_i, q_u, q_v	$s_i q_u q_v$	$s_i + q_u + q_v - 2$	$\max_w \{w(s_i + q_u + q_v - 2)\}$
0,0,0	0	– 2	$0\|_{w=0}$
0,0,1	0	– 1	$0\|_{w=0}$
0,1,0	0	– 1	$0\|_{w=0}$
0,1,1	0	0	$0\|_{w=0,1}$
1,0,0	0	– 1	$0\|_{w=0}$
1,0,1	0	0	$0\|_{w=0,1}$
1,1,0	0	0	$0\|_{w=0,1}$
1,1,1	1	1	$1\|_{w=1}$

Table 3 Triplet transformation into QUBO when $a > 0$

s_i, q_u, q_v	$s_i q_u q_v$	$s_i + q_u + q_v - 1$	$\min_w \{w(s_i + q_u + q_v - 1) + (s_i q_u + q_u q_v + q_v s_i) - (s_i + q_u + q_v) + 1\}$
0,0,0	0	– 1	$0\|_{w=1}$
0,0,1	0	0	$0\|_{w=0,1}$
0,1,0	0	0	$0\|_{w=0,1}$
0,1,1	0	1	$0\|_{w=0}$
1,0,0	0	0	$0\|_{w=0,1}$
1,0,1	0	1	$0\|_{w=0}$
1,1,0	0	1	$0\|_{w=0}$
1,1,1	1	2	$1\|_{w=0}$

Observe that columns 2 and 4 produce the same numbers in each table. Hence, the transformations are correct. In all of our QUBOs discussed so far, only H_C and H_E are not in the QUBO form with each involves 3 variables. Hence, we apply the transformation above to H_C and H_E to make it in the QUBO form. We note here that in general an interaction involving 4 or more variables can be reduced to pairwise by sequentially introducing new ancillary variables and reducing the degree of the interaction by 1 at each step. The middle column in Fig. 2 shows the total numbers of qubits for each of the QUBOs after the transformations. We note that for each of the triplets of qubits, one auxiliary qubit is needed to transform it into a QUBO.

The number of qubits can be further reduced by the following trick [36]. Suppose we want to encode the values $1, \ldots, N$. Define an integer m so that $2^m \leq N < 2^{m+1}$. Observe that $m = \lfloor log_2 N \rfloor$. In this case, we only need $m + 1$ binary variables instead of N binary variables to encode a variable which can take N values. It is easy to check that replacing

$$\sum_{n=1}^{N} n y_n \rightarrow \sum_{n=0}^{m-1} 2^n y_n + (N + 1 - 2^m) y_m$$

has the same effect. It is noted here that if $N \neq 2^{m+1} - 1$, the ground state may be degenerate, as the summation of $y's$ to a given integer is not always unique. Therefore, the number of qubits for each of AP, AC, and $AP - K$ can be reduced to $\lfloor \log_2(AP) \rfloor$, $\lfloor \log_2(AC) \rfloor$, and $\lfloor \log_2(AP - K) \rfloor$, respectively using this method. The last column in Fig. 2 shows the number of qubits after this technique has been applied.

We remark that each QUBO remains in the QUBO format after this qubit reduction since we replace a degree one expression with another degree one expression.

6 Experiments

This section presents experiments carried out to support the theoretical results. The objective of these experiments was to demonstrate that the theoretical results can actually be implemented. All experiments were performed on a classical laptop computer using D-Wave System's Ocean SDK which is a suite of Python tools to construct and solve problems in the form of Binary Quadratic Models (BQMs). To ensure correctness of the results, ExactSolver provided in the Ocean SDK within the dimod package was used. In our experiments, all main QUBOs were tested. These QUBOs are for the construction of rooted subtrees in Sect. 4.1.1, for weak defense in Sect. 4.2.1, for an attacking budget in Sect. 4.1.2, and for an optimal attack in Sect. 4.1.3. Observe that the QUBO for the optimal defense in Sect. 4.2.2 is the same as that for the optimal attack with an addition of H_d^∞. Hence, we only conducted experiments on H_d^∞ in Lemma 5.

QUBO	Before Reduction	After Reduction
$H_A + H_B$	$\lvert V\rvert$	$\lvert V\rvert$
H_w	$\lvert V\rvert + (2AC + 1)(1 + \lvert V\rvert^2)$	$\lvert V\rvert + (2\lfloor \log_2(AC)\rfloor + 1)(1 + \lvert V\rvert^2)$
H_{MB}	$\lvert V\rvert + AC(1 + \lvert V\rvert^2) + AP - K + 1$	$\lvert V\rvert + \lfloor \log_2(AC)\rfloor(1 + \lvert V\rvert^2) + \lfloor \log_2(AP - K)\rfloor + 1$
H_{attack}, H^i_{attack}	$\lvert V\rvert + AC(1 + \lvert V\rvert^2) + AP + (AC + 1)(1 + \lvert V\rvert^2)$	$\lvert V\rvert + \lfloor \log_2(AC)\rfloor(1 + \lvert V\rvert^2) + \lfloor \log_2(AP)\rfloor + (\lfloor \log_2(AC)\rfloor + 1)(1 + \lvert V\rvert^2)$
H^∞_d	d	d
$s_i H^i_{attack}$	$(2AC + 1)(1 + \lvert V\rvert^2) + \lvert V\rvert^2 + AP^2 + AP\lvert V\rvert + AC^2 - 2AC - 1 + (AC + 1)^2 + 3(AC + 1)\lvert V\rvert^2$	$(2\lfloor \log_2(AC)\rfloor + 1)(1 + \lvert V\rvert^2) + \lvert V\rvert^2 + \lfloor \log_2(AP)\rfloor^2 + \lfloor \log_2(AP)\rfloor\lvert V\rvert + \lfloor \log_2(AC)\rfloor^2 - 2\lfloor \log_2(AC)\rfloor - 1 + (\lfloor \log_2(AC)\rfloor + 1)^2 + 3(\lfloor \log_2(AC)\rfloor + 1)\lvert V\rvert^2$
$H_{defense}$	$(d - 1)((2AC + 1)(1 + \lvert V\rvert^2) + \lvert V\rvert^2 + AP^2 + AP\lvert V\rvert + AC^2 - 2AC - 1 + (AC + 1)^2 + 3(AC + 1)\lvert V\rvert^2) + d$	$(d - 1)((2\lfloor \log_2(AC)\rfloor + 1)(1 + \lvert V\rvert^2) + \lvert V\rvert^2 + \lfloor \log_2(AP)\rfloor^2 + \lfloor \log_2(AP)\rfloor\lvert V\rvert + \lfloor \log_2(AC)\rfloor^2 - 2\lfloor \log_2(AC)\rfloor - 1 + (\lfloor \log_2(AC)\rfloor + 1)^2 + 3(\lfloor \log_2(AC)\rfloor + 1)\lvert V\rvert^2) + d$

Fig. 2 Numbers of qubits before and after reduction

Table 4 Utility of each QUBO

QUBO	Utility
$H_A + H_B$	Create all possible rooted subtrees in a given tree $T = (V, E)$
H_W	Identify a part of a given security system prone to an attack
H_{MB}	Identify a minimum budget for a targeted total prize
H_{attack}	Compute a maximum total prize from a given budget
H^i_{attack}	Find a smaller total prize from an infinity cost placement
H^∞_d	Enforce exactly one placement of infinity cost
H_{defense}	Compute an optimal placement of infinity cost in a given system

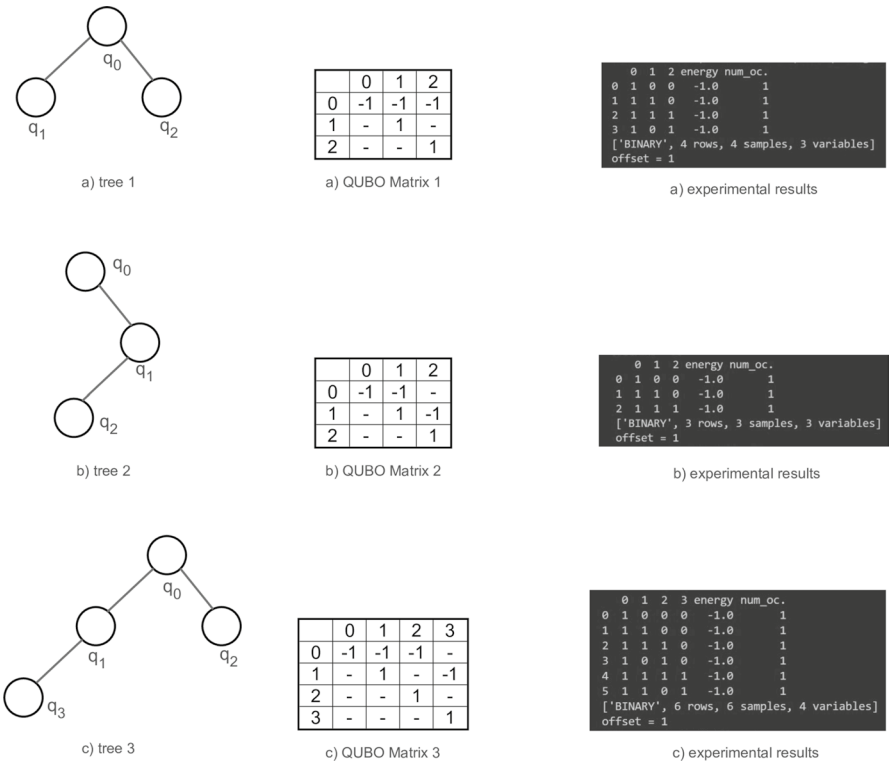


Fig. 3 Experimental results for the QUBO to generate subtrees

6.1 Attack problems

6.1.1 Rooted subtrees

This section discusses three experiments to illustrate the use of the QUBO in Lemma 2 in Sect. 4.1.1. In Fig. 3, the first column shows the trees corresponding to

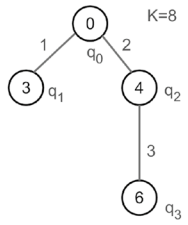
T in the cybersecurity model. The second and third columns show the corresponding QUBO matrices in the D-Wave annealer and the corresponding experimental results, respectively. The total energy of each QUBO in the experimental results can be computed by adding the offset of the QUBO to the energy shown.

In Fig. 3, (a) the experimental results show that there are 4 possible subtrees of tree 1. Each row represents the values of qubits 0, 1, and 2, the energy without offset, and the number of occurrences. Each subtree is identified by the qubits that are set to 1. The first subtree is a single node q_0. The second subtree is made up of two nodes q_0 and q_1. The third subtree consists of all nodes q_0, q_1, and q_2. The fourth subtree is made up of nodes q_0 and q_2. Subtrees of trees 2 and 3 in (b) and (c) are similarly identified. Hence, the total energies for trees 1, 2, and 3 are all 0's, i.e., -1 + 1, following the claim in Lemma 2.

6.1.2 Attacking budget

In this section, experiments for the problem of the attacking budget are discussed. Recall that in an attack on a security system an attacker may have a targeted total prize in mind and would like to know how much the budget is necessary for such an attack on the security system to obtain the targeted total prize. The formal problem definition was given in Definition 3.

Figure 4 displays two experiments for the QUBO for attacking budget. The same tree was used in both experiments. Only the targeted total prizes were different. In a) and b) the targeted total prizes were 8 and 7, respectively. Their corresponding QUBO matrices were shown in each case. In each case, the solutions are identified by the sum of the edge costs on the path in which the qubits q_0, q_1, q_2, and q_3 are all set to one. In a) the minimum budget required to obtain the total prize at least 8 is on the path $q_0 = q_2 = q_3 = 1$ and the sum of the edge costs is -645 + 650 = 5.



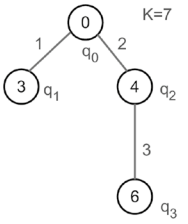
a) tree 1

	0	1	2	3	4	5	6	7
0	-10	-9	-8					
1		-380	240	360	-60	-120	-240	-480
2			-470	473	-80	-160	-320	-640
3				-590	-120	-240	-480	-960
4					170	40	80	160
5						360	160	320
6							800	640
7								1920

a) QUBO Matrix 1

```
   0  1  2  3  4  5  6  7 energy num_oc.
0  1  0  1  1  0  1  0  0 -645.0       1
['BINARY', 1 rows, 1 samples, 8 variables]
constant = 650
```

a) experimental results



b) tree 1

	0	1	2	3	4	5	6	7
0	-10	-9	-8					
1		-320	240	360	-60	-120	-240	-480
2			-390	473	-80	-160	-320	-640
3				-470	-120	-240	-480	-960
4					150	40	80	160
5						320	160	320
6							720	640
7								1760

b) QUBO Matrix 1

```
   0  1  2  3  4  5  6  7 energy num_oc.
0  1  1  1  0  0  0  0  0 -497.0       1
['BINARY', 1 rows, 1 samples, 8 variables]
constant = 500
```

b) experimental results

Fig. 4 Experimental results for the QUBO for attacking budget

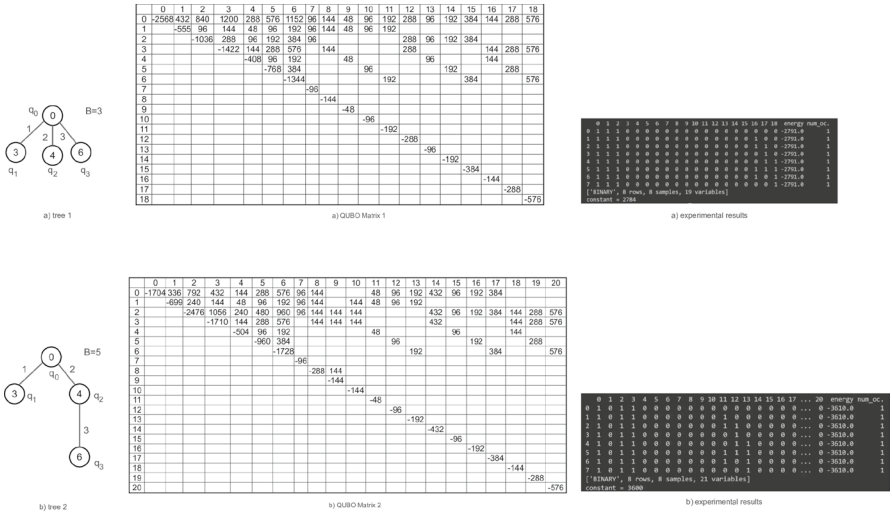


Fig. 5 Experimental results for the QUBO for the optimal attack

Similarly, in b) the minimum budget required to obtain the total prize at least 7 is on the paths $q_0 = q_1 = 1$ and $q_0 = q_2 = 1$ and the sum of the edge costs is -497 + 500 = 3. Hence, the experiments follow Theorem 1.

6.1.3 Optimal attack

In this section, experiments for the problem of the optimal attack are discussed. Recall that, given a security system and a budget, we want to compute the maximum total prize from an attack on the system. The formal problem definition was given in Definition 4. Two experiments were conducted and are shown in Fig. 5. In (a) tree 1 and (b) tree 2, the security systems comprise four nodes at different tree heights and budgets. The corresponding QUBO matrices are shown in the middle column and the experimental results are shown in the last column in the Figure. For tree 1, given a budget of 3, the experimental result shows that the maximum

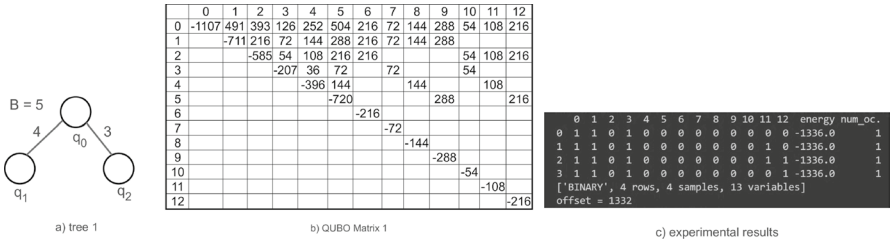


Fig. 6 Experimental results for the QUBO for weak defense

total prize from the optimal attack corresponding to the subtree $q_0 = q_1 = q_2 = 1$ is $-2791 + 2784 = -7$. For tree 2, given a budget of 5, the experimental result shows that the maximum total prize from the optimal attack corresponding to the subtree $q_0 = q_2 = q_3 = 1$ is -3610 + 3600 = -10. Hence, the experiments follow Theorem 2.

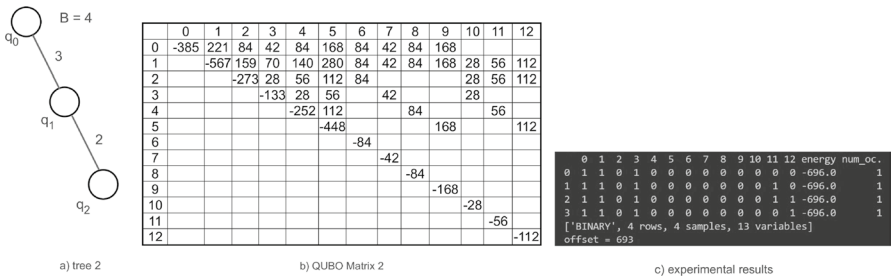
6.2 Defense problems

6.2.1 Weak defense

In this section, we have a security system $S = (c, p, T)$ associated with M and a budget B according to Definition 5. We would like to compute a subsystem of S that can be attacked with the highest possible cost, given a budget B. Three experiments were conducted as follows. In Fig. 6, (a) the security system S consists of three nodes, where the prizes are omitted for the sake of clarity, and the given budget B is 5. The corresponding QUBO matrix is shown in (b), and the experimental results are shown in (c). A solution is identified by the principal qubits q_0, q_1, and q_2 that are set to 1. In the experimental results, there are four possible solutions due to the different auxiliary qubit values, but all four solutions are identified with $q_0 = q_1 = 1$, which carries the corresponding highest possible cost of $-1336 + 1332 = -4$. Hence, the experiments follow Theorem 3.

In Fig. 7, (a) the security system S consists of three nodes, where the prizes are omitted for the sake of clarity, and the given budget B is 4. The corresponding QUBO matrix is shown in (b), and the experimental results are shown in (c). A solution is identified by the principal qubits q_0, q_1, and q_2 that are set to 1. In the experimental results, there are four possible solutions due to the different auxiliary qubit values, but all four solutions are identified with $q_0 = q_1 = 1$, which carries the corresponding highest possible cost of $-696 + 693 = -3$. Hence, the experiments follow Theorem 3.

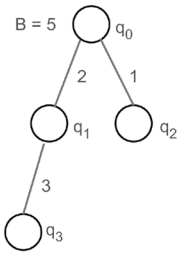
In Fig. 8, (a) the security system S consists of four nodes, where the prizes are omitted for the sake of clarity, and the given budget B is 5. The corresponding QUBO matrix is shown in (b), and the experimental results are shown in (c). A solution is identified by the principal qubits q_0, q_1, q_2 and q_3 that are set to 1. In the experimental results, there are eight possible solutions due to the different auxiliary qubit values, but all eight solutions are identified with $q_0 = q_1 = q_3 = 1$, which



	0	1	2	3	4	5	6	7	8	9	10	11	12
0	-385	221	84	42	84	168	84	42	84	168			
1		-567	159	70	140	280	84	42	84	168	28	56	112
2			-273	28	56	112	84				28	56	112
3				-133	28	56		42			28		
4					-252	112			84			56	
5						-448				168			112
6							-84						
7								-42					
8									-84				
9										-168			
10											-28		
11												-56	
12													-112

```
   0  1  2  3  4  5  6  7  8  9 10 11 12 energy num_oc.
0  1  1  0  1  0  0  0  0  0  0  0  0 -696.0       1
1  1  1  0  1  0  0  0  0  0  0  1  0 -696.0       1
2  1  1  0  1  0  0  0  0  0  0  1  1 -696.0       1
3  1  1  0  1  0  0  0  0  0  0  0  1 -696.0       1
['BINARY', 4 rows, 4 samples, 13 variables]
offset = 693
```

Fig. 7 Experimental results for the QUBO for weak defense



a) tree 3

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
0	-710	328	139	180	60	120	240	40	60			20	40	80	180	40	80	160			
1		-1030	100	437	100	200	400	40	60	60	60				180	40	80	160	60	120	240
2			-290	60	20	40	80	40	60		60	20	40	80							
3				-710	60	120	240		60	60	60				180				60	120	240
4					-210	40	80					20				40			60		
5						-400	160						40				80			120	
6							-720							80				160			240
7								-40													
8									-120	60											
9										-60											
10											-60										
11												-20									
12													-40								
13														-80							
14															-180						
15																-40					
16																	-80				
17																		-160			
18																			-60		
19																				-120	
20																					-240

b) QUBO Matrix 3

```
   0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 ... 20  energy num_oc.
0  1 1 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 ...  0 -1505.0       1
1  1 1 0 1 0 0 0 0 0 0 1 0 0 0 0 0 0 0 ...  0 -1505.0       1
2  1 1 0 1 0 0 0 0 0 0 1 1 0 0 0 0 0 0 ...  0 -1505.0       1
3  1 1 0 1 0 0 0 0 0 0 0 1 0 0 0 0 0 0 ...  0 -1505.0       1
4  1 1 0 1 0 0 0 0 0 0 0 1 1 0 0 0 0 0 ...  0 -1505.0       1
5  1 1 0 1 0 0 0 0 0 0 1 1 1 0 0 0 0 0 ...  0 -1505.0       1
6  1 1 0 1 0 0 0 0 0 0 1 0 1 0 0 0 0 0 ...  0 -1505.0       1
7  1 1 0 1 0 0 0 0 0 0 0 0 1 0 0 0 0 0 ...  0 -1505.0       1
['BINARY', 8 rows, 8 samples, 21 variables]
offset = 1500
```

c) experimental results

Fig. 8 Experimental results for the QUBO for weak defense

	0	1	2
0	-1	2	2
1		-1	2
2			-1

a) QUBO Matrix

```
   0  1  2 energy num_oc.
1  1  0  0   -1.0       1
3  0  1  0   -1.0       1
7  0  0  1   -1.0       1
0  0  0  0    0.0       1
2  1  1  0    0.0       1
4  0  1  1    0.0       1
6  1  0  1    0.0       1
5  1  1  1    3.0       1
['BINARY', 8 rows, 8 samples, 3 variables]
offset = 1
```

b) experimental results

Fig. 9 Experimental results for the QUBO for the exactly one placement

carries the corresponding highest possible cost of $-1505 + 1500 = -5$. Hence, the experiments follow Theorem 3.

6.2.2 Optimal defense

In the optimal defense, the goal is to find an edge to place an infinity cost so that the total prize from an attack is minimized. This problem was formally defined in Definition 7. Because the algorithm is hybrid and uses the QUBO H_{attack} in Sect. 4.1.3,

	0	1	2	3
0	-1	2	2	2
1		-1	2	2
2			-1	2
3				-1

a) QUBO Matrix

```
   0  1  2  3 energy num_oc.
0  1  0  0  0   -1.0       1
1  0  1  0  0   -1.0       1
2  0  0  1  0   -1.0       1
3  0  0  0  1   -1.0       1
['BINARY', 4 rows, 4 samples, 4 variables]
offset = 1
```

b) experimental results

Fig. 10 Experimental results for the QUBO for the exactly one placement

we therefore only conducted experiments with the QUBO H_d^∞ in Lemma 5 to show that it allows exactly one placement of infinity.

In Fig. 9, (a) is the QUBO matrix that represented H_3^∞. In (b) experimental results showed that there were exactly one $s_i = 1$. All these states are identified by the first 3 rows where the energies are -1 and the total energies are $-1 + 1 = 0$. This experiment therefore follows Lemma 5.

Similarly, in Fig. 10 (a) is the QUBO matrix that represented H_4^∞. In (b) experimental results showed that there were exactly one $s_i = 1$. All these states are identified by the four rows where the energies are -1 and the total energies are $-1 + 1 = 0$. This experiment therefore follows Lemma 5.

7 Discussion

In this research, a layered security model was defined, along with associated problems considered from both defensive and offensive perspectives. The objective was to demonstrate the construction of a Hamiltonian in the Quadratic Unconstrained Binary Optimization (QUBO) form for each of these problems. Under ideal conditions, it was shown—through both mathematical reasoning and experimental validation—that the proposed Hamiltonians exhibited behavior consistent with solving the problems. In other words, if these Hamiltonians are implemented on a quantum annealer, such as those developed by D-Wave Systems, solutions to the corresponding cybersecurity problems are expected. Table 4 presents a summary of the utility of each QUBO in the cybersecurity context discussed thus far. For each QUBO, the number of qubits required for its construction is identified in Fig. 2.

We emphasize here that, though our main goal of this article is to show how to formulate a correct QUBO in the context of cybersecurity, we do see the importance of the actual implementation of these Hamiltonians in a quantum annealer such as those of D-Wave systems.[1] We offer some opinions on this aspect as follows.

- In a practical situation, the number of logical qubits required for these Hamiltonians may be too large to implement in a current quantum annealer.

[1] At the time of this research Thailand does not have access to D-Wave computer.

- In the quantum annealing process, there exists a quantum phenomenon called *quantum tunneling* that helps avoid local minima and does not require an approximation to a wave function [6, 45]. Therefore, the quality of solutions is expected to be better than its classical annealing process.
- There has been ongoing debate whether quantum annealing is any faster than its classical algorithmic counterpart [46–54], due to the fact that if the problem has size N, it is usually the case that

$$T = O(\exp(\alpha N^{\beta})) \tag{2}$$

  In order for the system to remain in the ground state, for positive coefficients α and β, as $N \to \infty$. This is a consequence of the requirement that exponentially small energy gaps between the ground state of $H(t)$ and the first excited state, at some intermediate time, not lead to Landau–Zener transitions into excited states [49]. Even though most researchers do not believe that quantum annealing will give an exponential speedup over its classical algorithm on *NP*-complete problems and problems with similar hardness, the coefficients α and β may be smaller than those of known classical algorithms [36]. Hence, quantum annealing may still have some advantages on some classes of problems including the Optimal Attack Optimization Problem and the Infinity Placement Optimization Problem.
- In a practical implementation, the values of costs or prizes or budgets may be too large to be a constant in a quantum annealer. This is not a problem because we can scale these values down and scale the solution up to that of the original problem after the annealing process is completed.
- Problems in this article, particularly Optimal Attack Optimization Problem and Infinity Placement Optimization Problem, can potentially be applied in real life. A tree T can be a representation of an actual layered security network. A prize can be viewed as some valuable thing or information that is kept in the security network. A penetration cost can be viewed as a security measure or a security device. In real life, an expert would be needed to define prizes and costs for the model. Given this corresponding real-life configuration of a security network, a solution to the Optimal Attack Optimization Problem becomes an attack strategy that maximizes the total valuable thing/information and a solution to the Infinity Placement Optimization Problem becomes a defense strategy that indicates the portion of the security system that allows users to minimize the total asset an attacker can obtain.
- Optimization problems such as Optimal Attack Optimization Problem and Infinity Placement Optimization Problem may not be easily or efficiently mapped to D-Wave's Quadratic Unconstrained Binary Optimization (QUBO) or Ising model form. Complex constraints often require auxiliary variables, increasing embedding size and complexity.
- The current quantum hardware is not ready for real-world cybersecurity datasets. When the quantum hardware becomes ready, problems with real-world cybersecurity datasets such as intrusion detection should be tested.

8 Conclusion

In this article, the potential of quantum annealing to address critical cybersecurity challenges has been demonstrated by formulating both offensive and defensive scenarios as QUBO problems. Real-world cybersecurity concerns were translated into a quantum-ready framework through novel construction techniques and rigorous correctness proofs. The feasibility of these approaches was validated by experiments conducted using a D-Wave simulator, thereby paving the way for future research on scalable, quantum-assisted defense mechanisms. While the practical deployment of such methods remains dependent on the maturity of quantum hardware, the findings suggest that quantum annealing represents a promising complementary tool for enhancing cybersecurity in the quantum era.

Author contributions S.K. conceived the work, carried out all proofs, wrote the manuscript, and verified the experiments. J.K. wrote all the codes in the experiments and helped verify the paper.

Data availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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